

Explanation Stability and Trust Governance for Class-Incremental Network Intrusion Detection OFRA and ETG

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Abstract

Class-incremental network intrusion detection systems must learn new attack families without an oracle task identifier while retaining earlier decisions. This study examines silent explanation drift: per-class accuracy is preserved between consecutive checkpoints, but the features supporting the class score change materially. The event is operationalised using fixed per-class probes, a fixed Task-0 SHAP background, top-15 Jaccard below 0.70, and an accuracy change above -5 percentage points. OFRA uses a frozen two-layer MLP encoder, a rank-8 output-space LoRA adapter and binary head per family, DP-Means centroid memory, and bounded exemplars. Its class score is $s(f)=p(f)+0.5z(f)$, where $p(f)$ is the binary-head positive probability and $z(f)$ is the standardised negative nearest-centroid distance. Across five datasets with five complete runs per dataset, no isolated-path transition meets the silent-drift criterion, whereas the routed joint-score protocol records 6 events among 298 accuracy-preserved transitions (2.0%). Across four datasets with complete paired statistics, OFRA reduces forgetting by a geometric mean of 3.1-fold at an accuracy cost of 7.7-16.0 percentage points. Separately, ETG was evaluated independently on a replay-trained shared MLP. At seed 42, it certified 34 of 39 evaluable dataset-family cases, left five uncertified, recorded ten escalation events, did not emit two candidate explanation-drift notifications for uncertified cases, and recorded three re-certification events. Underlying NIDS alerts remained active by policy. The reported results exclude ongoing full-data experiments.

Keywords: class-incremental learning; network intrusion detection; explanation stability; low-rank adaptation; SHAP; explanation governance

1. Introduction

1.1 Background and Motivation

Network intrusion detection operates in a non-stationary environment in which attack families, traffic distributions, and operational constraints evolve over time. Retraining on all historical traffic is often impractical because of privacy, storage, and latency constraints. Class-incremental learning (Class-IL) instead learns classes sequentially and predicts among all observed classes without a task identifier at inference time (Masana et al., 2023; Wang et al., 2024). Its standard failure mode is catastrophic forgetting: performance on previously learned attacks deteriorates after later updates.

Predictive retention is necessary but insufficient for security operations. Analysts use feature attributions to prioritise alerts, refine rules, support forensic review, and justify automated actions. A class can retain its label while the evidence supporting that label changes after an update. An earlier explanation may then remain plausible but no longer describe the current score. Conventional accuracy and forgetting dashboards do not expose this condition.

1.2 Problem Statement

Continual NIDS research primarily measures predictive accuracy and forgetting, whereas explainable-AI research usually evaluates faithfulness or robustness at a static checkpoint. Neither perspective directly establishes whether explanations remain stable across successive task-agnostic Class-IL checkpoints. The research problem is therefore twofold: to measure explanation change without an all-data oracle or task identifier, and to determine how unfaithful or unstable explanations should be governed without suppressing the underlying detection alert.

1.3 Research Questions

RQ1: To what extent do model explanations drift across consecutive updates in task-agnostic class-incremental NIDS when predictive accuracy is preserved?

RQ2: What accuracy, forgetting, and parameter-cost trade-offs arise when oracle-free parameter isolation and routing are used to preserve prior knowledge and explanation stability?

RQ3: How can unfaithful or unstable explanations be governed without suppressing the underlying intrusion alert in an incremental NIDS?

1.4 Research Objectives

RO1: To develop and apply a consecutive-checkpoint protocol for quantifying silent explanation drift in task-agnostic class-incremental NIDS under accuracy-preserved conditions.

RO2: To design and evaluate OFRA in terms of forgetting, predictive accuracy, explanation stability, routing behaviour, and per-family parameter cost.

RO3: To design and evaluate a detector-agnostic ETG that converts explanation-faithfulness and stability evidence into certification, refusal, escalation, and re-certification actions while keeping the underlying intrusion alert active.

1.5 Contributions

- C1.** An oracle-free consecutive-checkpoint definition and measurement protocol for silent explanation drift.
- C2.** A fully specified OFRA implementation that separates a frozen family path from the routed cross-family joint score.
- C3.** A detector-agnostic ETG proof-of-concept state machine for explanation certification, refusal, withholding, escalation, and re-certification.
- C4.** A multi-dataset evaluation of forgetting, accuracy trade-offs, routed-score drift, and protocol limitations.

2. Literature Review

2.1 Class-Incremental Learning for Evolving Intrusion Detection

Continual learning updates a model from sequential data while retaining earlier knowledge; catastrophic forgetting remains its central failure mode (French, 1999; Wang et al., 2024). Class-IL is particularly demanding because prior samples may be unavailable and the model must distinguish all observed classes without a task identifier (Masana et al., 2023). Representative strategies include Elastic Weight Consolidation (Kirkpatrick et al., 2017), Learning without Forgetting (Li and Hoiem, 2017), and exemplar replay with prototype classification such as iCaRL (Rebuffi et al., 2017).

These concerns directly apply to NIDS, where new attack families emerge under severe imbalance and historical traffic may be difficult to retain. Existing work explores augmented replay (Amalapuram et al., 2022), strategic selection and forgetting control (Zhang et al., 2024), and clustering-based memory with distillation and class weighting (Guo et al., 2025). These studies primarily evaluate predictive metrics rather than longitudinal explanation continuity.

2.2 Explainability, Faithfulness, and Stability

SHAP provides an additive framework for attributing model outputs to input features (Lundberg and Lee, 2017). NIDS studies compare SHAP, LIME, Integrated Gradients, and DeepLIFT using faithfulness, complexity, robustness, reliability, and analyst relevance (Kalakoti et al., 2025). Explanations may change under small perturbations even when the class is unchanged (Ghorbani et al., 2019), motivating attribution-robustness methods (Wang and Kong, 2024) and deletion tests. Most work nevertheless examines static checkpoints or perturbed inputs rather than sequential updates.

2.3 Parameter Isolation and Continual Low-Rank Adaptation

LoRA freezes a base model and expresses an update through low-rank trainable matrices (Hu et al., 2022). CL-LoRA applies shared and task-specific adapters to reduce interference in rehearsal-free Class-IL (He et

al., 2025). Frozen backbones, modular adapters, and routed experts can improve retention and efficiency, but the cited literature does not evaluate tabular NIDS explanation stability after both routing and cross-family score competition are included.

2.4 Research Gap

The reviewed literature provides limited evidence on explanation continuity between consecutive task-agnostic NIDS checkpoints and limited guidance on converting explanation evidence into persistent actions. This study addresses that intersection while distinguishing a structurally fixed family path, a routed joint score, and a separate governance proof of concept.

3. Methodology

3.1 Experimental Scope and Metric Definitions

The study contains two experimental pipelines. Pipeline A evaluates shared-parameter Class-IL baselines and OFRA, including isolated-path and routed joint-score stability. Pipeline B evaluates an ETG proof of concept independently on a replay-trained shared MLP; its results characterise the state machine rather than an integrated OFRA-ETG system. In the predictive benchmark, the accuracy matrix is task-level: $A[i,j]$ is task- j accuracy after increment i . Reported forgetting is the mean, over old tasks, of the reduction from each task's best prior accuracy to its final accuracy. Silent drift uses per-class accuracy between adjacent checkpoints and is defined separately below.

3.2 OFRA Architecture and Training

OFRA uses a two-layer MLP encoder $d \rightarrow 128 \rightarrow 128$ with ReLU activations. It is supervised-pretrained on Task 0 for eight epochs and then frozen. Each family receives an output-space LoRA adapter with rank 8 and alpha 16, followed by a 128-to-2 binary classifier. The module contains 2,048 LoRA parameters plus 258 classifier parameters, or 2,306 trainable parameters per family. Each new family is trained for ten epochs using Adam (learning rate $1e-3$, batch 256), at most four negatives per positive, and 50 farthest-first exemplars per old family. Families with fewer than 1,000 positives use focal loss (gamma=2, positive-class alpha=0.75); larger families use cross-entropy.

For family f , $p(f)$ is the positive-class softmax probability of its binary head. DP-Means stores at most 32 centroids per family, with its distance threshold initialised from the 0.30 quantile of sampled pairwise embedding distances. Let $z(f)$ be the within-sample z-score of the negative nearest-centroid distance across current families. Inference predicts $\text{argmax}_f s(f)$, where $s(f) = 1.0p(f) + 0.5z(f)$. DP-Means is therefore not a hard pre-selector, and no separate learned calibration model is used.

OFRA architecture and inference procedure

All family heads compete through a joint score; DP-Means is not a hard pre-selector.

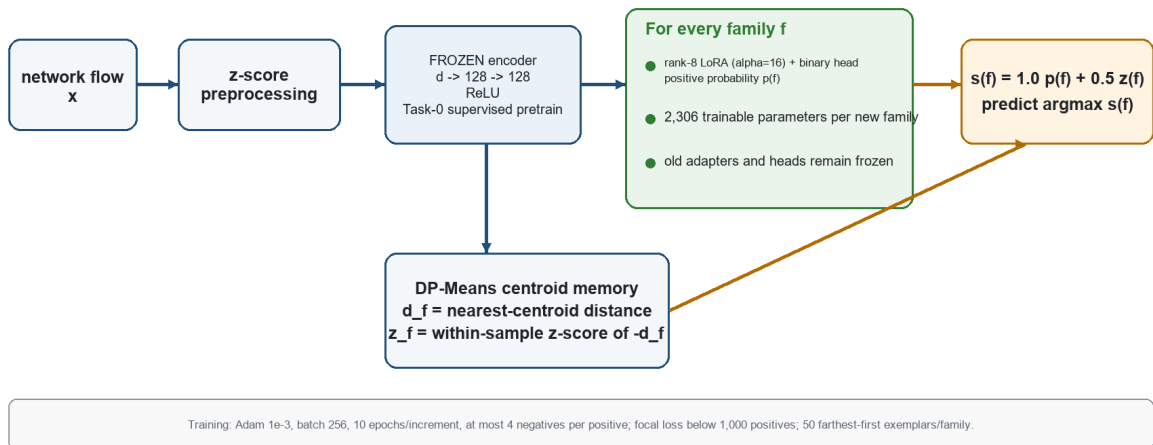


Figure 1. OFRA implementation. All binary family heads and router distances contribute to the joint cross-family score.

3.3 Oracle-Free Explanation-Stability Protocol

For each class and seed, a fixed probe of at most 200 test samples is reused at every checkpoint. SHAP GradientExplainer uses a fixed background of at most 100 Task-0 training samples and 200 SHAP samples. The class attribution vector is mean absolute SHAP over the probe. Consecutive vectors are compared by cosine similarity, Kendall tau, and top-15 Jaccard overlap. A transition is accuracy-preserved when $A(c,t) - A(c,t-1) > -0.05$ and is silent drift when its top-15 Jaccard is below 0.70. The isolated-path explainer covers normalisation, frozen encoder, and target-family logit difference while excluding routing. The routed analysis explains the continuous target-family joint score; it does not explain the discontinuous argmax operation.

$$\text{SilentDrift}(c,t)=1 \text{ when } A(c,t)-A(c,t-1) > -0.05 \text{ and } J15(a(c,t-1),a(c,t)) < 0.70.$$

A separate deletion-faithfulness test replaces SHAP-selected features with the fixed Task-0 background mean and compares the score reduction with five random feature sets. The sensitivity analysis varies the accuracy, Jaccard, and Kendall thresholds while holding $k=15$ fixed. The completed analysis does not include a separate decomposition of model-update effects and data-distribution shift.

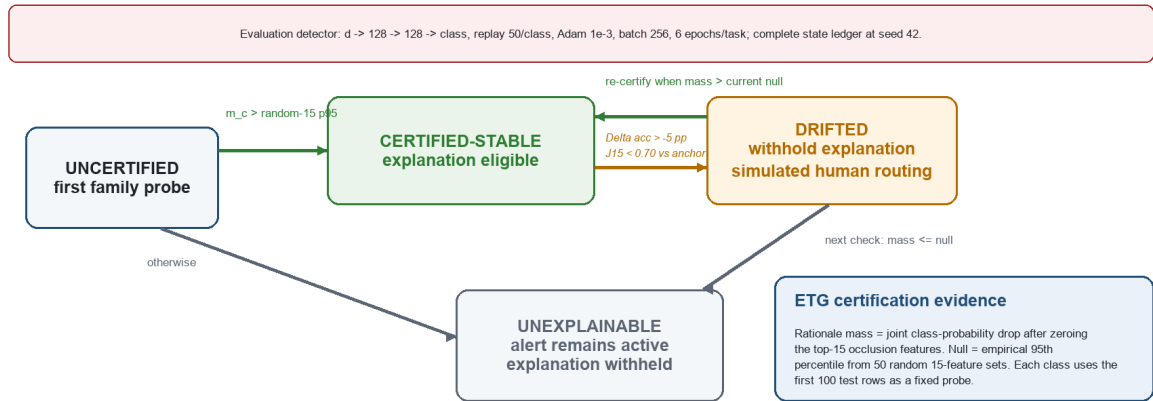
3.4 ETG Proof-of-Concept Mechanism and Scope

ETG is evaluated separately on a shared $d \rightarrow 128 \rightarrow 128 \rightarrow \text{class}$ MLP trained with 50 replay samples per class, Adam at $1e-3$, batch 256, and six epochs per task. The complete ledger uses seed 42. Each class uses the first 100 test rows as a fixed probe. Occlusion importance is the class-probability change when one standardised feature is set to zero. Rationale mass is the joint probability drop after zeroing the top-15 occlusion features; the admission null is the empirical 95th percentile from 50 random 15-feature sets. A family is CERTIFIED-STABLE when mass exceeds the null and otherwise becomes UNEXPLAINABLE.

A certified family becomes DRIFTED when probe accuracy changes by more than -5 percentage points and Jaccard against its frozen admission rationale is below 0.70. The explanation is withheld and human routing is simulated; the detection alert remains active. For an UNEXPLAINABLE family, the same condition is treated only as a candidate explanation-drift signal: no drift-governance notification is emitted because no certified reference rationale exists, while the underlying detection alert remains active. Under the implemented rule, re-certification checks only whether current rationale mass exceeds the current null; restored Jaccard stability is not rechecked.

ETG proof-of-concept trust state machine

Scope: separate shared-MLP + replay experiment; not yet integrated with OFRA's routed score.



Re-certification currently evaluates rationale mass only; restored Jaccard stability is not re-evaluated.

Figure 2. Detector-agnostic ETG proof-of-concept state machine. The reported ledger is separate from OFRA.

3.5 Datasets, Comparators, Seeds, and Statistical Scope

NSL-KDD and UNSW-NB15 use their documented train/test files. CIC-IDS-2017 caps BENIGN at 200,000; CIC-IDS-2018 and NF-ToN-IoT-v2 use approximately 50,000 majority samples per class while retaining available rare samples, followed by a fixed stratified 80/20 split. Processed model-input dimensions, rather

than raw field counts, are reported in Table 1. The reported experiments use the capped datasets listed in Table 1 and do not include CIC-IoT-2023 or the uncapped full-data runs.

Table 1. Dataset sizes and processed input dimensions used by the reported capped-data protocol.

Dataset	Used records	Effective families	Model input dim	Split
NSL-KDD	148,517	5	122	Official train/test
UNSW-NB15	257,673	10	196	Official train/test
CIC-IDS-2017	754,376 / 2.83M	7*	78	Capped + fixed 80/20
CIC-IDS-2018	300,928	7	78	Capped + fixed 80/20
NF-ToN-IoT-v2	377,916	10	39	Capped + fixed 80/20

* The CIC-IDS-2017 cache contains one unused WebAttack label slot. Reported metrics exclude that slot and use seven non-empty effective families.

The formal OFRA explanation runs use seeds $\{0,1,2,3,42\}$. Data caps, splits, and task order are fixed; seeds affect stochastic training, negative/exemplar selection, and explanation probes rather than creating five independent data splits. The benchmark includes Naive, EWC, LwF, Replay, Replay-DPMeans, iCaRL, NCM-Frozen, and OFRA; the paired analysis reported here covers a configuration-matched four-method comparison subset. Paired t-tests are Holm-corrected across four forgetting comparisons. Baseline replay budgets are 200 examples per class versus OFRA's 50, so the comparison is not memory-budget matched.

3.6 Implementation and Reproducibility Boundaries

The preprocessing first fits StandardScaler on the complete training partition and then fits OFRA's internal feature statistics on the same partition before the task stream. No test labels are used, but the procedure incorporates unlabeled distributional information from later training-task classes. This introduces training-distribution look-ahead and represents a transductive benchmark rather than a deployment-clean online protocol. The public snapshot contains the result files and consolidated evidence summary but does not include exact package versions, device metadata, or dataset checksums.

4. Results

4.1 RQ1: Explanation Drift Across Consecutive Accuracy-Preserving Updates

Preserved predictive accuracy does not imply explanation stability. The archived summaries for the shared-parameter baselines report silent-explanation-drift rates of 55.0% for Naive, 37.8% for Replay, and 45.6% for Replay-DPMeans. Under OFRA, the isolated family path records no silent-drift events across 299 accuracy-preserving transitions (0/299), whereas the routed joint-score protocol records six events across 298 accuracy-preserving transitions (6/298, 2.0%). Across the five datasets, the OFRA routed joint-score event rate ranges from 0.0% to 6.25%. These results show that isolated-path stability does not extend automatically to the routed inference process, where cross-family score competition and checkpoint-dependent standardisation remain relevant.

No isolated-path transition met the silent-drift criterion, although mean top-15 Jaccard similarity across datasets ranged from 0.980 to 0.997. The stored routed-score wrapper and the NumPy inference implementation use different standard-deviation conventions, so the 6/298 result is not a bit-exact reproduction of deployed scoring. It characterises the stored experimental protocol rather than an estimate of operational prevalence.

Table 2. Archived silent-explanation-drift results for the shared-parameter baselines and OFRA; the OFRA results comprise complete runs over five datasets and five random seeds.

Dataset	Naive	Replay	Replay-DPMeans	OFRA isolated path	OFRA routed joint score
NSL-KDD	52.9%	15.4%	28.0%	0.0%	2/36 (5.6%)
UNSW-NB15	61.2%	36.7%	42.9%	0.0%	0/80 (0.0%)
CIC-IDS-2017	78.6%	82.4%	84.6%	0.0%	3/48 (6.25%)

Dataset	Naive	Replay	Replay-DPMeans	OFRA isolated path	OFRA routed joint score
CIC-IDS-2018	100%	80.6%	92.1%	0.0%	0/51 (0.0%)
NF-ToN-IoT-v2	16.7%	2.9%	5.9%	0.0%	1/83 (1.2%)
Pooled	55.0%	37.8%	45.6%	0/299 (0.0%)	6/298 (2.0%)

4.2 RQ2: Retention, Accuracy, and Parameter-Cost Trade-offs of Oracle-Free Isolation and Routing

Across the four datasets with auditable five-seed paired results, OFRA achieves a geometric-mean forgetting reduction of 3.07-fold. The paired t-tests for NSL-KDD, UNSW-NB15, and CIC-IDS-2017 remain statistically significant after Holm correction, whereas the result for CIC-IDS-2018 is not significant. Relative to the corresponding comparators, OFRA incurs mean accuracy reductions of 7.7 to 16.0 percentage points. At this operating point, lower forgetting is accompanied by lower predictive accuracy.

The paired analysis is limited to methods with protocol-matched result files. NF-ToN-IoT-v2 is excluded from Table 3 because its available five-seed estimate-a 2.33-fold forgetting reduction relative to iCaRL-was produced under a legacy Transformer protocol with $d=64$ and five masked-feature pretraining epochs. The present study uses the $d>128 \rightarrow 128$ MLP OFRA protocol with eight epochs of supervised Task-0 pretraining, so the two estimates are not directly comparable.

Table 3. Five-seed forgetting means $\pm$ standard deviations, paired t-tests, and Holm correction within the four-test forgetting family.

Dataset	Comparator	OFRA forgetting	Comparator forgetting	Reduction / Holm-adjusted p	OFRA accuracy change
NSL-KDD	Replay-DPMeans	0.0188 $\pm$ 0.0111	0.1236 $\pm$ 0.0359	6.59x / 0.0085	-16.0 pp
UNSW-NB15	iCaRL	0.0999 $\pm$ 0.0123	0.3525 $\pm$ 0.0111	3.53x / 8.36e-6	-7.7 pp
CIC-IDS-2017	Replay-DPMeans	0.0194 $\pm$ 0.0069	0.0565 $\pm$ 0.0156	2.91x / 0.0267	-10.6 pp
CIC-IDS-2018	iCaRL	0.0902 $\pm$ 0.0427	0.1181 $\pm$ 0.0192	1.31x / 0.1019 n.s.	-7.8 pp
Geometric mean	Four comparisons			3.07x	

Note: OFRA accuracy change is defined as OFRA accuracy minus comparator accuracy; negative values therefore indicate lower mean accuracy for OFRA.

OFRA adds 2,306 trainable adapter-and-head parameters per attack family. It also stores up to 32 DP-Means router centroids and 50 representative samples per family, neither of which is included in the 2,306-parameter count. In addition, the baseline replay methods retain up to 200 samples per class, whereas OFRA retains 50, meaning that the comparison does not match the complete memory budgets.

Across 555 checkpoint-class observations, the oracle-family-access diagnostic shows a mean gap of 22.2 percentage points relative to multiclass accuracy. The gap aggregates effects from routing, binary-head competition, and cross-family score comparability; it is not a measure of router cost alone. The measurements cover forgetting, accuracy, and per-family trainable-parameter count. Total memory, runtime, explanation latency, and energy consumption were not measured.

4.3 RQ3: Governing Unfaithful or Unstable Explanations While Preserving Intrusion Alerts

The ETG proof of concept demonstrates that explanation-trust states can be separated from the underlying intrusion-detection decision at the policy level. The complete seed-42 ledger certifies 34 of 39 evaluable attack families at admission and refuses explanation certification to the remaining five. It records ten CERTIFIED-STABLE-to-DRIFTED escalation events across nine unique dataset-family pairs, three re-certification events based only on rationale mass, and no demotions.

For families that failed admission, two candidate explanation-drift alarms were not emitted because no certified reference rationale was available. Under the ETG policy, this gating applies only to the untrusted explanation and the associated explanation-drift governance signal; it does not suppress the underlying NIDS intrusion alert. In the DRIFTED state, the current explanation is withheld and human escalation is simulated.

In the UNEXPLAINABLE state, the underlying detection alert remains active, but no uncertified explanation or explanation-based drift alarm is issued.

These seed-42 results count ETG state transitions on a separate shared MLP; they do not measure ground-truth drift, false-positive rates, analyst outcomes, or ETG responses to OFRA's six routed-score events. Alert preservation is an ETG policy property and has not been tested end to end.

Table 4. Seed-42 governance outcomes from the detector-agnostic ETG proof of concept; the 'not emitted' column refers to explanation-drift governance alarms, not underlying intrusion-detection alerts.

Dataset	Certified at admission	Not certified	Explanation-drift alarms not emitted	Escalation events	Re-certifications
NSL-KDD	5/5	0	0	2	1
UNSW-NB15	9/10	1	1	1	0
CIC-IDS-2017	6/7	1	0	2	1
CIC-IDS-2018	7/7	0	0	4	1
NF-ToN-IoT-v2	7/10	3	1	1	0
Total	34/39	5	2	10	3

A separate five-seed experiment repeats only the admission stage. The certified-family counts are 5.0 +/- 0.0 for NSL-KDD, 9.2 +/- 0.4 for UNSW-NB15, 6.2 +/- 0.75 for the evaluable CIC-IDS-2017 families, 7.0 +/- 0.0 for CIC-IDS-2018, and 7.0 +/- 0.0 for NF-ToN-IoT-v2. Replacing occlusion ranking with input gradients changes 19 of the 39 admission decisions, showing that the certification outcome depends materially on the attribution method.

Under the evaluated repair protocol, rare-family performance fell by approximately 45 percentage points for UNSW-NB15 Shellcode and 14 percentage points for CIC-IDS-2017 Infiltration, cautioning against automatic repair without further validation.

5. Discussion

The results distinguish structural stability from operational stability. No silent-drift event was observed on the isolated path after OFRA's encoder and old family modules were frozen, but the set of competing families and router-score normalisation continue to change. The routed joint-score protocol therefore retains 2.0% silent drift. OFRA's retention benefit also carries a material accuracy cost and a diagnostic gap under oracle family access. OFRA thus occupies a retention-accuracy trade-off rather than dominating the comparators.

ETG provides a policy interface for certifying, withholding, and escalating explanations. Integrating OFRA's routed score with the ETG ledger remains future work.

6. Limitations and Threats to Validity

Several limitations affect the interpretation of these results. Zero silent-drift events apply only to OFRA's isolated family path; the routed analysis explains a continuous score rather than the discrete argmax, and its standardisation is not bit-exact with deployed inference. The analysis does not include a separate model-update versus data-distribution-shift decomposition, and all threshold scans keep $k=15$ fixed. Preprocessing uses statistics from the complete training partition before the stream, introducing training-distribution look-ahead. Memory budgets also differ between the baselines and OFRA. With only five paired runs, the smallest attainable two-sided exact Wilcoxon signed-rank p-value is 0.0625 when all five non-zero differences have the same sign. The small-sample paired t-test results, including the Holm-adjusted values, are therefore interpreted cautiously.

ETG was evaluated independently of OFRA, and complete state transitions were measured at one seed. Re-certification checks rationale mass but not recovered Jaccard stability, while the same class-level probes are reused for admission and monitoring. The simulated human-routing action has not been evaluated with analysts. Three datasets cap majority classes, and the CIC-IDS-2017 label mapping remains unresolved. The

experiments do not report exact environment versions, device metadata, dataset checksums, runtime, explanation latency, per-alert false-positive rates, or analyst burden.

7. Conclusion and Future Work

The experiments support three conclusions within the evaluated protocol. First, explanation continuity cannot be inferred from accuracy preservation, and consecutive-checkpoint monitoring can identify silent explanation drift without an all-data reference model. Second, no isolated-path transition met the silent-drift criterion, whereas the routed joint score recorded 6/298 events; the forgetting reduction was accompanied by a 7.7-16.0 percentage-point accuracy cost across the four complete comparisons. Third, ETG executed certification, withholding, escalation, and re-certification on a separate shared MLP; its integration with OFRA remains unevaluated.

Future work will first connect ETG to OFRA's exact routed score and fit preprocessing statistics using only data available at each increment. Further experiments will strengthen re-certification, vary top-k, separate model-update effects from data-distribution shift, rebuild the CIC-IDS-2017 label mapping, validate uncapped-data results, and measure latency, computational cost, and analyst workload.

Code and Data Availability

Code, result files, and the consolidated evidence summary are available at <https://github.com/1838904818/Class-IL>. Public datasets remain subject to their original licences and distribution terms.

Appendix A. Reproducibility Checklist

Table 5. Implementation and reproducibility summary.

Item	Reported configuration or limitation
OFRA training	Task-0 supervised pretrain 8 epochs; incremental heads 10 epochs; Adam 1e-3; batch 256; fixed data splits.
Per-family module	Output-space LoRA rank 8, alpha 16, binary 128-to-2 head; 2,306 trainable parameters/family.
Router/inference	Up to 32 DP-Means centroids/family; $s(f)=p(f)+0.5z(-\text{nearest distance})$; no learned calibration model.
Explanation protocol	GradientExplainer; ≤ 200 test probes/class; ≤ 100 Task-0 background rows; 200 SHAP samples; mean absolute SHAP; top-15.
OFRA evidence	Five datasets x seeds {0,1,2,3,42}; 25 complete routed joint-score runs; 6/298 silent-drift events.
ETG scope	Separate shared-MLP+replay PoC; complete ledger seed 42; five-seed robustness covers admission only.
Statistics	Four complete paired t-tests; Holm correction within forgetting family; comparator set narrower than full benchmark.
Unreported or unresolved items	Stream-only preprocessing without later-class look-ahead; exact software and device versions; dataset checksums; runtime and latency; CIC17 label remapping; and validated full-data replications.

References

- Amalapuram, S. K., Tamma, B. R., and Channappayya, S. S. (2022). Augmented memory replay-based continual learning approaches for network intrusion detection. *IEEE INFOCOM Workshops*, 1-6.
- DeYoung, J., Jain, S., Rajani, N. F., Lehman, E., Xiong, C., Socher, R., and Wallace, B. C. (2020). ERASER: A benchmark to evaluate rationalized NLP models. *Proceedings of ACL*.
- French, R. M. (1999). Catastrophic forgetting in connectionist networks. *Trends in Cognitive Sciences*, 3(4), 128-135.
- Ghorbani, A., Abid, A., and Zou, J. (2019). Interpretation of neural networks is fragile. *Proceedings of AAAI*.
- Guo, C., Li, X., Cheng, J., Yang, S., and Gong, H. (2025). Continual learning for intrusion detection under evolving network threats. *Future Internet*, 17(10), 456.
- He, J., Duan, Z., and Zhu, F. (2025). CL-LoRA: Continual low-rank adaptation for rehearsal-free class-incremental learning. *Proceedings of CVPR*, 30534-30544.
- Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Y., Li, Y., Wang, S., Wang, L., and Chen, W. (2022). LoRA: Low-rank adaptation of large language models. *International Conference on Learning Representations*.
- Kalakoti, R., Vaarandi, R., Bahsi, H., and Nomm, S. (2025). Evaluating explainable AI for deep learning-based network intrusion detection system alert classification. *arXiv:2506.07882*.
- Kirkpatrick, J., Pascanu, R., Rabinowitz, N., et al. (2017). Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, 114(13), 3521-3526.
- Kulis, B., and Jordan, M. I. (2012). Revisiting k-means: New algorithms via Bayesian nonparametrics. *arXiv:1111.0352*.
- Li, Z., and Hoiem, D. (2017). Learning without forgetting. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 40(12), 2935-2947.

- Lundberg, S. M., and Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*.
- Masana, M., Liu, X., Twardowski, B., Menta, M., Bagdanov, A. D., and van de Weijer, J. (2023). Class-incremental learning: Survey and performance evaluation on image classification. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(5), 5513-5533.
- Moustafa, N., and Slay, J. (2015). UNSW-NB15: A comprehensive data set for network intrusion detection systems. *Military Communications and Information Systems Conference*, 1-6.
- Rainio, O., Teuvo, J., and Klen, R. (2024). Evaluation metrics and statistical tests for machine learning. *Scientific Reports*, 14(1), 6086.
- Rebuffi, S.-A., Kolesnikov, A., Sperl, G., and Lampert, C. H. (2017). iCaRL: Incremental classifier and representation learning. *Proceedings of CVPR*, 2001-2010.
- Sarhan, M., Layeghy, S., Moustafa, N., and Portmann, M. (2021). NetFlow datasets for machine learning-based network intrusion detection systems. *Big Data Technologies and Applications*.
- Sharafaldin, I., Lashkari, A. H., and Ghorbani, A. A. (2018). Toward generating a new intrusion detection dataset and intrusion traffic characterization. *ICISSP*, 108-116.
- Tavallaei, M., Bagheri, E., Lu, W., and Ghorbani, A. A. (2009). A detailed analysis of the KDD CUP 99 data set. *IEEE Symposium on Computational Intelligence for Security and Defense Applications*, 1-6.
- Wang, F., and Kong, A. W.-K. (2024). Certified l2 attribution robustness via uniformly smoothed attributions. *arXiv:2405.06361*.
- Wang, L., Zhang, X., Su, H., and Zhu, J. (2024). A comprehensive survey of continual learning: Theory, method and application. *IEEE Transactions on Pattern Analysis and Machine Intelligence*.
- Zhang, X., Zhao, R., Jiang, Z., Chen, H., Ding, Y., Ngai, E. C. H., and Yang, S.-H. (2024). Continual learning with strategic selection and forgetting for network intrusion detection. *arXiv:2412.16264*.